Question 7. How do you pass-by-reference?

Answer:

```
return_type function_name(T & identifier);
```

The identifier is now an alias for the argument.

Question 8. How do you pass a read only argument by reference?

Answer:

```
return_type function_name(const T & identifier);
```

Once you define a parameter as `const`, you will not be able to modify it in the function.

Question 9. What are the important differences between using a pointer and a reference?

Answer: Several differences between using a pointer and a reference are:

• A pointer can be re-assigned any number of times, while a reference cannot be reassigned after initialization.

• A pointer can point to NULL (`nullptr` in C++11), while a reference can never be referred to NULL.

• It is not possible to take the address of a reference as it is done with pointers.

• There is no reference arithmetic.

Question 10. How do you set a default value for a parameter?

Answer:

```
return_type function_name(T identifier = rvalue)
```

The parameters with default value must be placed at the end of the parameter list.

Question 11. How do you create a template function?

Answer:

```
template<class T>
return_type function_name(parameter_list);

template<typename T>
return_type function_name(parameter_list);
```

Note that the parameter type can be specified, when calling the function, explicitly or implicitly. Also note that there is no technical difference between using `class` or `typename` besides code readability (`typename` for primitive types and `class` for classes).

Example:

```
template<typename T>
T temp_sum(T a, T b) {return a+b;}

struct Processor{
    int a;
    int apply(int b) {return a+b;}
};

template<class T>
int temp_sum_2(int a, int b) {
    T processor;
    processor.a = a;
    return processor.apply(b);
}

int main(){
    //implicit, foo equals 3
    int foo = temp_sum(1,2);

    //explicit, bar equals 3
    int bar = temp_sum_2<Processor>(1,2);
}
```

Question 12. How do you declare a pointer to a function?

Answer:

```
return_type (*identifier) (list_parameter_types)
```

Example:

```
int my_sum(int a, int b) {return a+b;}
int main(){
    int(*p_func)(int,int);
    p_func = & my_sum;

    // foo equals 3
    int foo = p_func(1,2);
}
```

Question 13. How do you prevent the compiler from doing an implicit conversion with your class?

Answer: Use the keyword `explicit` to define the constructor:

```
explicit Classname(parameter_list)
```

Question 14. Describe all the uses of the keyword `static` in C++.

Answer: Inside a function, using the keyword `static` means that once the variable has been initialized it remains in memory up until the end of the program.

Inside a class definition, either for a variable or for a member function, using the keyword `static` means that the there is only one copy of them per class, and shared between instances.

As a global variable in a file of code, using the keyword `static` means that the variable is private within the scope of the file.

Question 15. Can a static member function be const?

Answer: When the `const` qualifier is applied to a non-static member function it implies that member function can not change the instance class when called (i.e. can not change any non `mutable` members from `*this`). Since static member function are defined at a class level, where there is no notion of `this` the `const` qualifier for member functions does not apply.

Question 16. C++ constructors support the initialization of member data via an initializer list. When is this preferable to initialization inside the body of the constructor?

Answer: The initialization list has to be used for const members, references and with members without default constructors, but for any type of members initialization through the initialization list is still preferable, since it is for efficient. Using the initialization list, the members are initialized calling directly their constructors. If the initialization is done in the body of the constructor for each member being initialized there is an instance of it created and then a copy assignment operation is called to assign that instance to its respective member.

Question 17. What is a copy constructor, and how can the default copy constructor cause problems when you have
pointer data members?

Answer: A copy constructor allows you to create a new object as a copy of an existing instance. The default copy constructor creates the new object by copying the existing object, member by member, and thus when there are member pointer you end up with two objects pointing to the same object.

It is important to note that the copy constructor is called every time a function receives an object via the pass-by-value mechanism. This means that the copy constructor needs to be implemented using a pass by reference. Otherwise you will be recursively calling the copy constructor. You should always set the parameter for a copy constructor to be const.

```
ClassName( const ClassName& other );
ClassName( ClassName& other );
ClassName( volatile const ClassName& other);
ClassName( volatile ClassName& other );
```

Question 18. What is the output of the following code:

```
#include <iostream>
using namespace std;

class A
{
public:
    int * ptr;
    ~A()
    {
        delete(ptr);
    }
};

void foo(A object_input)
{
    ;
}

int main()
{
    A aa;
    aa.ptr = new int(2);
```

```
    foo(aa);
    cout<<(*aa.ptr)<<endl;
    return 0;
}
```

Answer: The output of the code is an uncertain number, depending on the compiler used; for some compilers it could generate an error. The reason for this is that we do not define our own copy constructor.

When we call the foo function, the compiler will generate a default copy constructor which will shallow copy every data members defined in class A. This will lead to the result that two pointers, one in temporary object and the other in the object aa, will point to the same area in memory. When we get out the foo function, the compiler will automatically call the destructor function of the temporary object in which the pointer will be deleting and the area it points to will be free. In this situation, the pointer in aa will still point to the same area which has been free. When we try to visit the data through the pointer in aa, we will get garbage information.

Question 19. How do you overload an operator?

Answer:

```
type operator symbol (parameter_list);
```

If you define the operator outside of the class, then it will be a global operator function.

Example:

```
struct FooClass{int a;};
int operator + (FooClass lhs, FooClass rhs) {
    return lhs.a + rhs.a;
}
```

Question 20. What are smart pointers?

Answer: A smart pointer is a class built to mimic a pointer (offering dereferencing, indirection, arithmetic) that also offers extra features to simplify the usage, sharing and management of resources.

C++11 comes with three implementations of smart pointers: **shared_ptr**, **unique_ptr**, and **weak_ptr**.

Example:

```
//shared_pointer maintains
//a reference count
//when the count is zero the object
//pointed to is destroyed
std::shared_ptr<int> foo(new int(3));
std::shared_ptr<int> bar = foo;

//memory not released
//bar is still in scope
foo.reset();

//releases the memory,
//since no one is using it
bar.reset();
}
```

Question 21. What is encapsulation?

Answer: Encapsulation is the ability to expose an interface while hiding implementation. This is usually achieved through access modifiers (public, private, protected, etc.).

Question 22. What is a polymorphism?

Answer: Polymorphism is the ability for a set of classes to all be referenced through a common interface.

Question 23. What is inheritance?

Answer: Inheritance is the ability for one class to extend another through sub-classing. This is also referred to as "white-box" (the opposite of "black-box") re–use. A library can provide base classes that may be extended by the application developer.

Question 24. What is a virtual function? What is a pure virtual function and when do you use it?

Answer: Virtual functions are functions that are resolved by the compiler, at runtime, to the most derived version with the same signature. This means that if a function that was defined using a base class `Foo`, with a virtual member function `f`, is called using an instance of a sub class `FooChild`, that function is going to be dynamically binded to the implementation of the sub class (regardless that the actual code only refers to the base class).

A pure virtual function is a virtual function with no implementation in the base class, making the base class abstract (and thus can't be instantiated). Derived classes are forced to override the pure virtual function if they want to be instantiated. You use the same syntax as the virtual function but add `=0` to its declaration within the class.

Question 25. Why are virtual functions used for destructors? Can they be used for constructors?

Answer: Destructors are recommended to be defined as virtual, so the proper destructor (in the class hierarchy) is called at running time.

When calling a constructor, the caller needs to know the exact type of the object to be created, and thus they cannot be virtual.

Question 26. Write a function that computes the factorial of a positive integer.

Answer:

```
//for implementation
int factorial(int n){
    int output =1;
    for (int i =2 ; i <= n ; ++i)
        output *= i;
    return output;
}

//recursive implementation
int factorial(int n){
   if (n == 0) return 1;
   return n*factorial(n-1);
}

//tail recursive implementation
int factorial(int n, int last = 1){
    if (n == 0) return last;
    return factorial(n-1, last * n);
}
```

Question 27. Write a function that takes an array and returns the subarray with the largest sum.

Answer:

```
#include <vector>
#include <algorithm> // std::max

using namespace std;

template <typename T>
T max_sub_array(vector<T> const & numbers){
    T max_ending = 0, max_so_far = 0;
    for(auto & number: numbers){
        max_ending = max(0, max_ending + number);
```

```
        max_so_far = max(max_so_far, max_ending);
    }
    return max_so_far;
}
```

Question 28. Write a function that returns the prime factors of a positive integer.

Answer:

```
#include <vector>
using namepsace std;

vector<int> prime_factors(int n){
    vector<int> factors;
    for (int i = 2; i <= n/i ; ++i)
        while (n % i == 0) {
            factors.push_back(i);
            n /= i;
        }
    if (n > 1)
        factors.push_back(n);
    return factors;
}
```

Question 29. Write a function that takes a 64-bit integer and swaps the bits at indices `i` and `j`.

Answer:

```
long swap_bits (long x, const int &i, const int &j){
    if ( ((x >>i) & 1L) != ((x>>j) & 1L) )
        x ^= (1L <<i ) | (1L <<j);
    return x;
}
```

Question 30. Write a function that reverses a single linked list.

Answer:

```
#include <memory> // shared_ptr
using namespace std;

template<typename T>
struct node_t {
    T data;
    shared_ptr<node_T<T>> next;
};

//recursive implementation
template<typename T> shared_ptr<node_t<T>>
    reverse_linked_list(
        const shared_ptr<node_t <T>> &head){
    if (!head || !head->next) {
        return head;
    }

    shared_ptr<node_t<T>>
      new_head = reverse_linked_list(head->next);
    head->next->next = head;
    head->next = nullptr;
    return new_head;
}

//while implementation
template<typename T> shared_ptr<node_t<T>>
    reverse_linked_list(
        const shared_ptr<node_t <T>> &head){
    shared_ptr<node_t<T>>
        prev = nullptr, curr = head;
     while(curr) {
  shared_ptr<node_t<T>> temp = curr-> next;
  curr->next = prev;
  prev = curr;
  curr = temp;
```

```
    }
    return prev;
}
```

Question 31. Write a function that takes a string and returns true if its parenthesis are balanced.

Answer:

```
#include<string>
#include<stack>
using namespace std;

bool is_par_balanced(const string input)
{
    //"())())"=> false
    //"(a(dd)()(()))"=>true
    stack<char> par_stack;
    for(auto &c: input)
    {
        if(c==')')
        {
            if(par_stack.empty())
            return false;
            else if(par_stack.top()=='(')
            par_stack.pop();
        }
        else if(c=='(')
        par_stack.push(c);
    }
    return par_stack.empty();
}
```

Question 32. Write a function that returns the height of an arbitrary binary tree.

Answer:

```
#include<memory> //std::shared_ptr
#include<algorithm> //std::max
using namespace std;

template <typename T>
struct BinaryTree {
    T data;
    shared_ptr<BinaryTree<T>> left, right;
};


template <typename T>
int height(
    const shared_ptr<BinaryTree<T>> &tree,
    int count = -1){
        if (!tree) return count;
        return max(
            height(tree->left, count + 1),
            height(tree->right, count +1));
}
```

3.5 Monte Carlo simulations. Numerical methods.

Question 1. How would you compute π using Monte Carlo simulations? What is the standard deviation of this method?

Answer: An Acceptance–Rejection type method can be used to approximate π as follows: generate N points uniformly in the $[-1,1] \times [-1,1]$ square and accept a point (x,y) if the point is in the unit disk $D(0,1)$, i.e., if $x^2 + y^2 \leq 1$. If the number of accepted points is A, then the ratio $\frac{A}{N}$ converges in the limit as $N \to \infty$ to the ratio of the area of the unit disk to the area of the square, which is equal to $\frac{\pi}{4}$. Therefore,

$$\pi \approx \frac{4A}{N}.$$

The standard deviation of the method is $O\left(\frac{1}{\sqrt{N}}\right)$. We make this more precise by computing below the coefficient of $\frac{1}{\sqrt{N}}$ in $O\left(\frac{1}{\sqrt{N}}\right)$; see (3.79).

Let $U_1, U_2, \cdots$ be a sequence of independent identically distributed bivariate random variables uniformly distributed in $[-1,1] \times [-1,1]$. Denote by $\mathbf{1}_{D(0,1)}$ the indicator function of the unit disk $D(0,1)$, i.e.,

$$\mathbf{1}_{D(0,1)}(x,y) = \begin{cases} 1, & \text{if } (x,y) \in D(0,1); \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$X_i = \mathbf{1}_{D(0,1)}(U_i), \quad \forall\, i \geq 1.$$

The random variables X_i, $i \geq 1$, are independent and identically distributed, with

$$\begin{aligned} E[X_i] &= E\left[\mathbf{1}_{D(0,1)}(U_i)\right] = \iint\limits_{D(0,1)} \frac{1}{4} dx dy \\ &= \frac{\pi}{4}. \end{aligned} \tag{3.77}$$

Since X_i, $i \geq 1$, are integrable, it follows from the strong law of large numbers that

$$\lim_{n \to \infty} \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{\pi}{4} \quad \text{almost surely.}$$

Note that $X_1 + X_2 + \cdots + X_n$ counts how many points out of the randomly selected n points reside in the disk $D(0,1)$. Thus, for N large enough,

$$\frac{X_1 + X_2 + \cdots + X_N}{N} \approx \frac{\pi}{4}. \tag{3.78}$$

To calculate the variance of the estimation in (3.78), note that, for $1 \leq i \leq N$,

$$\left(\mathbf{1}_{D(0,1)}(U_i)\right)^2 = \mathbf{1}_{D(0,1)}(U_i), \quad \forall\, 1 \leq i \leq N,$$

and therefore, using (3.77), we find that

$$\begin{aligned} \operatorname{var}(X_i) &= E[X_i^2] - (E[X_i])^2 \\ &= E\left[\left(\mathbf{1}_{D(0,1)}(U_i)\right)^2\right] - \left(\frac{\pi}{4}\right)^2 \\ &= E\left[\mathbf{1}_{D(0,1)}(U_i)\right] - \frac{\pi^2}{16} \\ &= \frac{\pi}{4} - \frac{\pi^2}{16} \\ &= \frac{\pi(4-\pi)}{16}. \end{aligned}$$

Then,

$$\begin{aligned}
& \operatorname{var}\left(\frac{X_1 + X_2 + \cdots + X_N}{N}\right) \\
= \quad & \frac{1}{N^2}\left(\operatorname{var}(X_1) + \cdots + \operatorname{var}(X_N)\right) \\
= \quad & \frac{1}{N^2} \cdot N \, \frac{\pi(4-\pi)}{16} \\
= \quad & \frac{\pi(4-\pi)}{16N}.
\end{aligned}$$

By taking the square root of the variance, we conclude that the standard deviation of this Monte Carlo method for estimating π is

$$\frac{\sqrt{\pi(4-\pi)}}{4} \, \frac{1}{\sqrt{N}} \approx 0.82 \, \frac{1}{\sqrt{N}}. \qquad (3.79)$$

Question 2. What methods do you know for generating independent samples of the standard normal distribution?

Answer: The three most commonly used methods to generate independent standard normal samples are:

- Box–Muller (using the Marsaglia–Bray algorithm in order to avoid estimating trigonometric functions);
- Acceptance–Rejection;
- Inverse Transform.

For details on these methods, see Glasserman [2]. □

Question 3. How do you generate a geometric Brownian motion stock path using random numbers from a normal distribution?

Answer: Consider a stock whose price follows the geometric Brownian motion

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \qquad (3.80)$$

where μ and σ are the drift and the volatility of the stock price, and W_t is a Weiner process. To generate a price path for the stock between time 0 and time T, discretize the time interval into m equal time steps of size $\delta t = \frac{T}{m}$, and let $t_j = j\delta t$, for $j = 0 : m$.

By integrating (3.80) between t_j and t_{j+1}, it follows that

$$S_{t_{j+1}} - S_{t_j} \;=\; \mu \int_{t_j}^{t_{j+1}} S_t dt \;+\; \sigma \int_{t_j}^{t_{j+1}} S_t dW_t. \quad (3.81)$$

We use the following approximations:

$$\begin{aligned} \int_{t_j}^{t_{j+1}} S_t dt &\approx S_{t_j}(t_{j+1} - t_j) \\ &= S_{t_j}\delta t; \qquad (3.82) \\ \int_{t_j}^{t_{j+1}} S_t dW_t &\approx S_{t_j}(W_{t_{j+1}} - W_{t_j}) \\ &= S_{t_j}\sqrt{\delta t}\, Z_{j+1}, \qquad (3.83) \end{aligned}$$

where Z_{j+1} is a standard normal variable, since W_t is a Wiener process and therefore $W_{t_{j+1}} - W_{t_j}$ is a normal variable with mean 0 and variance $t_{j+1} - t_j = \delta t$, i.e.,

$$W_{t_{j+1}} - W_{t_j} \;=\; \sqrt{\delta t} Z_{j+1}. \quad (3.84)$$

Note that Z_j, for $j = 1 : m$, are independent standard normals.

If $z_1, z_2, \ldots, z_m$ are independent samples of the standard normal distribution, we obtain from (3.81–3.83) that (3.80) can be discretized as follows:

$$S_{t_{j+1}} - S_{t_j} \;=\; \mu S_{t_j}\delta t \;+\; \sigma S_{t_j}\sqrt{\delta t} z_{j+1},$$

for $j = 0 : (m - 1)$, which can be written as

$$S_{t_{j+1}} \;=\; S_{t_j}\left(1 + \mu\delta t + \sigma\sqrt{\delta t} z_{j+1}\right),$$

for $j = 0 : (m-1)$.

Note that there is a very small probability that the price path above becomes negative, which is a drawback of using this discretization.

A price path which is always positive can be generated using Itô's formula to express (3.80) as

$$d(\ln(S_t)) \;=\; \left(\mu - \frac{\sigma^2}{2}\right) dt \;+\; \sigma dW_t. \tag{3.85}$$

By integrating (3.85) between t_j and t_{j+1}, it follows that

$$\begin{aligned} \ln(S_{t_{j+1}}) - \ln(S_{t_j}) \;&=\; \ln\left(\frac{S_{t_{j+1}}}{S_{t_j}}\right) \\ &=\; \left(\mu - \frac{\sigma^2}{2}\right)(t_{j+1} - t_j) \;+\; \sigma(W_{t_{j+1}} - W_{t_j}) \\ &=\; \left(\mu - \frac{\sigma^2}{2}\right)\delta t \;+\; \sigma\sqrt{\delta t}Z_{j+1}, \end{aligned} \tag{3.86}$$

for $j = 0 : (m-1)$, where (3.84) was used for (3.86).

Then, (3.86) can be discretized as follows:

$$\ln\left(\frac{S_{t_{j+1}}}{S_{t_j}}\right) \;=\; \left(\mu - \frac{\sigma^2}{2}\right)\delta t + \sigma\sqrt{\delta t}z_{j+1},$$

for $j = 0 : (m-1)$, and therefore

$$S_{t_{j+1}} \;=\; S_{t_j}\,\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)\delta t + \sigma\sqrt{\delta t}z_{j+1}\right),$$

for all $j = 0 : (m-1)$. $\square$

Question 4. How do you generate a sample of the standard normal distribution from 12 independent samples of the uniform distribution on $[0, 1]$?

Answer: If $u_1, u_2, \ldots, u_{12}$ are 12 independent samples of the uniform distribution on $[0, 1]$, then

$$\sum_{i=1}^{12} u_i \;-\; 6 \tag{3.87}$$

can be used as a sample of the standard normal distribution.

To see this, recall from the Central Limit Theorem that, if X_i, $i \geq 1$, are independent identically distributed random variables with finite expected value $E[X]$ and standard deviation $\sigma(X)$, then

$$\lim_{n\to\infty} \frac{\frac{1}{n}\left(\sum_{i=1}^n X_i\right) \; - \; E[X]}{\frac{\sigma(X)}{\sqrt{n}}} \; = \; Z, \tag{3.88}$$

where Z is the standard normal distribution and the convergence in (3.88) is in distribution.

Let $U_1, U_2, \ldots, U_{12}$ be 12 independent uniform distributions on $[0,\ 1]$. Using (3.88) for $n = 12$ and $X_i = U_i$, $i = 1:12$, we infer that

$$Z \; \approx \; \frac{\frac{1}{12}\left(\sum_{i=1}^{12} U_i\right) \; - \; E[U]}{\frac{\sigma(U)}{\sqrt{12}}}, \tag{3.89}$$

where

$$\begin{aligned} E[U] &= \int_0^1 u \, du \; = \; \frac{1}{2}; \qquad (3.90) \\ \sigma^2(U) &= E[U^2] - (E[U])^2 = \int_0^1 u^2 du - \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{3} - \frac{1}{4} \; = \; \frac{1}{12}, \end{aligned}$$

and thus

$$\sigma(U) \; = \; \frac{1}{\sqrt{12}}. \tag{3.91}$$

From (3.89–3.91), we obtain that

$$\begin{aligned} Z &\approx \frac{\frac{1}{12}\left(\sum_{i=1}^{12} U_i\right) \; - \; \frac{1}{2}}{\frac{1}{12}} \\ &= \sum_{i=1}^{12} U_i \; - \; 6, \end{aligned}$$

and therefore (3.87) can be used as an approximate sample of the standard normal distribution.

Note that this is an inefficient method, since it uses 12 samples from the uniform distribution to generate one approximate sample of the standard normal distribution, and all the samples that it generates are in the interval $[-6, 6]$. More efficient methods for generating independent standard normal samples are Box–Muller, which uses two uniform distribution samples to generate two samples of the standard normal distribution, and Acceptance–Rejection. □

Question 5. What is the rate of convergence for Monte Carlo methods?

Answer: If n is the number of paths in the Monte Carlo simulation and m is the number of time steps between 0 and T used in the discretization of each path, then the convergence rate of the Monte Carlo simulation is

$$O\left(\max\left(\frac{T}{m}, \frac{1}{\sqrt{n}}\right)\right).$$

The estimate above holds for Monte Carlo simulations on multi asset derivative securities, i.e., is independent of the number of underlying assets of the derivative security, unlike finite differences and numerical integration methods, where the convergence slows down as the number of underlying assets increases. □

Question 6. What variance reduction techniques do you know?

Answer: The variance reduction techniques are used to reduce the constant factor corresponding to the Monte Carlo approximation error $O\left(\frac{1}{\sqrt{n}}\right)$. Some of the most commonly used variance reduction techniques are:

- Control Variates;

• Antithetic Variables;

• Moment Matching.

For details on these methods and their implementation, see Glasserman [2]. □

Question 7. How do you generate samples of normal random variables with correlation ρ?

Answer: Assume that you can generate two samples z_1 and z_2 from two independent standard normal variables Z_1 and Z_2. Let $X_1 = Z_1$ and $X_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_2$. Note that X_2 is a linear combination of independent normal variables, and therefore X_2 is a normal variable as well. Also,

$$\begin{aligned}
\operatorname{corr}(X_1, X_2) &= \operatorname{corr}(Z_1, \rho Z_1 + \sqrt{1-\rho^2} Z_2) \\
&= \rho \operatorname{corr}(Z_1, Z_1) \\
&\quad + \sqrt{1-\rho^2} \operatorname{corr}(Z_1, Z_2) \\
&= \rho \operatorname{var}(Z_1) \\
&= \rho,
\end{aligned}$$

since $\operatorname{var}(Z_1) = 1$ and $\operatorname{corr}(Z_1, Z_2) = 0$, since Z_1 and Z_2 are independent.

We conclude that X_1 and X_2 are normal random variables with correlation ρ. Thus, starting with two independent standard normal samples z_1 and z_2,

$$x_1 = z_1 \quad \text{and} \quad x_2 = \rho z_1 + \sqrt{1-\rho^2} z_2$$

are samples of normal random variables with correlation ρ. □

Question 8. What is the order of convergence of the Newton's method?

Answer: If it is convergent, Newton's method is quadratically (second order) convergent.

Recall that, given an initial guess x_0, the Newton's method recursion for solving $f(x) = 0$, where $f : \mathbb{R} \rightarrow \mathbb{R}$, is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad \forall\, k \geq 0. \qquad (3.92)$$

The quadratic convergence of Newton's method can be stated formally as follows: Let x^* be a solution of $f(x) = 0$. If $f(x)$ is a twice differentiable function with $f''(x)$ continuous, if $f'(x^*) \neq 0$, and if x_0 is close enough to x^*, then there exists $M > 0$ and n_M a positive integer such that

$$\frac{|x_{k+1} - x^*|}{|x_k - x^*|^2} < M, \quad \forall\, k \geq n_M. \qquad (3.93)$$

To provide the intuition behind (3.93), note that, since $f(x^*) = 0$, the recursion (3.92) can be written as

$$\begin{aligned} x_{k+1} - x^* & \\ &= x_k - x^* - \frac{f(x_k) - f(x^*)}{f'(x_k)} \\ &= \frac{f(x^*) - f(x_k) + (x_k - x^*) f'(x_k)}{f'(x_k)}. \qquad (3.94) \end{aligned}$$

Recall from the linear Taylor expansion of the function $f(x)$ around the point x_k that, if $f''(x)$ is continuous, then there exists a point c_k between x^* and x_k such that

$$\begin{aligned} f(x^*) &= f(x_k) + (x^* - x_k) f'(x_k) \\ &\quad + \frac{(x^* - x_k)^2}{2} f''(c_k). \qquad (3.95) \end{aligned}$$

From (3.94) and (3.95), we find that

$$x_{k+1} - x^* = (x^* - x_k)^2\, \frac{f''(c_k)}{2f'(x_k)},$$

and conclude that

$$\frac{|x_{k+1} - x^*|}{|x_k - x^*|^2} = \left| \frac{f''(c_k)}{2f'(x_k)} \right|. \qquad (3.96)$$

If $f''(x)$ and $f'(x)$ are continuous functions such that $f'(x^*) \neq 0$, it follows that, if x_k is close to x^*, then $\left|\frac{f''(c_k)}{2f'(x_k)}\right|$ is close to $\left|\frac{f''(x^*)}{2f'(x^*)}\right| < \infty$. Therefore, the term $\left|\frac{f''(c_k)}{2f'(x_k)}\right|$ is uniformly bounded if x_k is close enough to x^*, and (3.96) can be written formally as (3.93). □

Question 9. Which finite difference method corresponds to trinomial trees?

Answer: Forward Euler. As an explicit finite difference method, the Forward Euler discretization of the Black–Scholes PDE gives the finite difference value of the option at a node as a linear combination of the option values at three nodes on the prior time step, which is similar to the risk neutral formula for trinomial trees.

If the calibration of the trinomial trees is done in the log space, i.e., if the up and down factors are calibrated to the normal distribution of $\ln(S)$, and if the Forward Euler discretization is done for the constant coefficients PDE obtained by the change of variables $x = \ln(S)$, the classical trinomial tree recursion and the Forward Euler recursion are almost identical. □

3.6 Probability. Stochastic calculus.

Question 1. What is the exponential distribution? What are the mean and the variance of the exponential distribution?

Answer: The density function of the exponential random variable X with parameter $\alpha > 0$ is

$$f(x) = \begin{cases} \alpha\, e^{-\alpha x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

The expected value and the variance of the exponential random variable X are

$$E[X] = \frac{1}{\alpha} \quad \text{and} \quad \mathrm{var}(X) = \frac{1}{\alpha^2}.$$

To see this, use integration by parts to find that

$$\begin{aligned} \int_0^\infty x e^{-\alpha x} dx &= -\frac{x e^{-\alpha x}}{\alpha}\bigg|_0^\infty + \frac{1}{\alpha}\int_0^\infty e^{-\alpha x} dx \\ &= 0 - \frac{e^{-\alpha x}}{\alpha^2}\bigg|_0^\infty = \frac{1}{\alpha^2}; \\ \int_0^\infty x^2 e^{-\alpha x} dx &= -\frac{x^2 e^{-\alpha x}}{\alpha}\bigg|_0^\infty + \frac{2}{\alpha}\int_0^\infty x e^{-\alpha x} dx \\ &= 0 + \frac{2}{\alpha} \times \frac{1}{\alpha^2} = \frac{2}{\alpha^3}. \end{aligned}$$

Then,

$$E[X] = \int_{-\infty}^\infty x f(x)\, dx = \alpha \int_0^\infty x e^{-\alpha x}\, dx = \frac{1}{\alpha};$$

$$E[X^2] = \int_{-\infty}^\infty x^2 f(x)\, dx = \alpha \int_0^\infty x^2 e^{-\alpha x}\, dx = \frac{2}{\alpha^2}.$$

Therefore,

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = \frac{2}{\alpha^2} - \left(\frac{1}{\alpha}\right)^2 = \frac{1}{\alpha^2}. \quad \square$$

Question 2. If X and Y are independent exponential random variables with mean 6 and 8, respectively, what is the probability that Y is greater than X?

Answer: The probability density functions of X and Y are, respectively:

$$f_X(x) = \begin{cases} \frac{1}{6}e^{-\frac{x}{6}}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0; \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{8}e^{-\frac{y}{8}}, & \text{if } y \geq 0; \\ 0, & \text{if } y < 0. \end{cases}$$

Since X and Y are independent, the joint probability density function $f_{XY}(x,y)$ of (X,Y) is the product of the marginal probability density functions, i.e.,

$$\begin{aligned} f_{XY}(x,y) &= f_X(x)\, f_Y(y) \\ &= \begin{cases} \frac{1}{48}e^{-\frac{x}{6}-\frac{y}{8}}, & \text{if } x \geq 0,\ y \geq 0; \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Let

$$A = \{(x,y) \in \mathbb{R}^2 : y \geq x\}.$$

The probability that Y is greater than X can be found by evaluating the double integral

$$\begin{aligned} P(Y \geq X) &= \iint_A f_{XY}(x,y)dxdy \\ &= \frac{1}{48}\int_0^\infty \int_0^y e^{-\frac{x}{6}-\frac{y}{8}}dxdy \end{aligned}$$

as follows:

$$\begin{aligned}
P(Y \geq X) &= \frac{1}{48}\int_0^\infty e^{-\frac{y}{8}} \left(\int_0^y e^{-\frac{x}{6}} dx \right) dy \\
&= \frac{1}{48}\int_0^\infty e^{-\frac{y}{8}} \left(-6e^{-\frac{x}{6}} \Big|_0^y \right) dy \\
&= \frac{1}{8}\int_0^\infty e^{-\frac{y}{8}} \left(1 - e^{-\frac{y}{6}} \right) dy \\
&= \frac{1}{8}\int_0^\infty \left(e^{-\frac{y}{8}} - e^{-\frac{7}{24}y} \right) dy \\
&= \frac{1}{8}\left(8 - \frac{24}{7} \right) \\
&= \frac{4}{7}. \quad \square
\end{aligned}$$

Question 3. What are the expected value and the variance of the Poisson distribution?

Answer: A Poisson distribution is a random variable X taking nonnegative integer values with probabilities

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \ \forall\, k \geq 0,$$

where $\lambda > 0$ is a fixed positive number.

We show that the expected value and the variance of the Poisson distribution X are

$$E[X] = \lambda \quad \text{and} \quad \mathrm{var}(X) = \lambda.$$

By definition,

$$\begin{aligned}
E[X] &= \sum_{k=0}^{\infty} P(X = k) \cdot k = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!}\, k \\
&= e^{-\lambda}\lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}. \qquad (3.97)
\end{aligned}$$

Since the Taylor series expansion for e^t is

$$e^t = \sum_{k=0}^{\infty} \frac{t^k}{k!},$$

it follows that

$$\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = e^{\lambda}; \tag{3.98}$$

$$\sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = e^{\lambda}. \tag{3.99}$$

From (3.97) and (3.98), we find that $E[X] = \lambda$.

To calculate $\mathrm{var}(X)$, note that

$$\begin{aligned} E[X^2] &= \sum_{k=0}^{\infty} P(X=k) \cdot k^2 \\ &= \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} k^2 \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \frac{k\lambda^k}{(k-1)!}, \end{aligned}$$

which can be written as

$$\begin{aligned} E[X^2] &= e^{-\lambda} \sum_{k=1}^{\infty} \frac{(k-1)\lambda^k}{(k-1)!} + e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} \\ &= e^{-\lambda}\lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} + e^{-\lambda}\lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda^2 + \lambda, \end{aligned}$$

where (3.98) and (3.99) were used for the last equality.

We conclude that

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = \lambda. \quad \square$$

Question 4. A point is chosen uniformly from the unit disk. What is the expected value of the distance between the point and the center of the disk?

Answer: The expected value of the distance between a uniformly chosen point in the unit disk D and the center of the disk can be computed as $E\left[\sqrt{X^2+Y^2}\right]$, where (X,Y) is uniformly distributed in the unit disk D. The probability density function of (X,Y) is

$$f(x,y) = \begin{cases} \frac{1}{\pi}, & \text{if } x \in D; \\ 0, & \text{otherwise.} \end{cases}$$

Then,

$$E\left[\sqrt{X^2+Y^2}\right] = \frac{1}{\pi}\iint_D \sqrt{x^2+y^2}\,dxdy. \qquad (3.100)$$

Using the polar coordinates substitution $x = r\cos\theta$ and $y = r\sin\theta$, with $0 \le r \le 1$ and $0 \le \theta < 2\pi$, and recalling that $dxdy = rdrd\theta$, we obtain from (3.100) that

$$\begin{aligned} & E\left[\sqrt{X^2+Y^2}\right] \\ = \ & \frac{1}{\pi}\int_0^{2\pi}\int_0^1 \sqrt{r^2\left(\cos^2\theta+\sin^2\theta\right)}\, rdrd\theta \\ = \ & \frac{1}{\pi}\int_0^1\int_0^{2\pi} r^2 d\theta\, dr \qquad (3.101) \\ = \ & \frac{1}{\pi}\int_0^1 2\pi r^2\, dr \ = \ 2\int_0^1 r^2 dr \\ = \ & \frac{2}{3}, \end{aligned}$$

where for (3.101) we used the fact that $\cos^2\theta + \sin^2\theta = 1$ for all θ. □

Question 5. Consider two random variables X and Y with mean 0 and variance 1, and with joint normal distribution. If $\text{cov}(X, Y) = \frac{1}{\sqrt{2}}$, what is the conditional probability $P(X > 0|Y < 0)$?

Answer: From the definition of conditional probability, it follows that

$$P(X > 0|Y < 0) = \frac{P(X > 0, Y < 0)}{P(Y < 0)}. \tag{3.102}$$

Note that

$$P(Y < 0) = \frac{1}{2}, \tag{3.103}$$

since Y is a standard normal random variable.

In order to compute $P(X > 0, Y < 0)$, let

$$W = \sqrt{2}\, X - Y. \tag{3.104}$$

Since $E[X] = E[Y] = 0$, it follows that $E[W] = 0$. Moreover, since

$$\text{var}(X) = \text{var}(Y) = 1 \text{ and } \text{cov}(X, Y) = \frac{1}{\sqrt{2}},$$

we obtain that

$$\begin{aligned} \text{var}(W) &= \text{var}\left(\sqrt{2}\, X - Y\right) \\ &= \text{var}\left(\sqrt{2}\, X\right) - 2\, \text{cov}\left(\sqrt{2}X, Y\right) + \text{var}(Y) \\ &= 2\text{var}(X) - 2\sqrt{2}\, \text{cov}(X, Y) + \text{var}(Y) \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} \text{cov}(W, Y) &= \text{cov}\left(\sqrt{2}\, X - Y, Y\right) \\ &= \sqrt{2}\, \text{cov}(X, Y) - \text{var}(Y) \\ &= 0. \end{aligned}$$

Note that $W = \sqrt{2}\, X - Y$ is a normal random variable since X and Y have joint normal distribution. Moreover, since $E[W] = 0$, $\mathrm{var}(W) = 1$, and $\mathrm{cov}(W, Y) = 0$, it follows that W and Y are independent standard normal variables.

From (3.104), we find that

$$X = \frac{1}{\sqrt{2}}(W + Y).$$

Then, the probability of the event $\{X > 0, Y < 0\}$ can be written as

$$\begin{aligned} & P(X > 0, Y < 0) \\ = \; & P\left(\frac{1}{\sqrt{2}}(W + Y) > 0, Y < 0\right) \\ = \; & P(W + Y > 0, Y < 0). \end{aligned} \tag{3.105}$$

The two straight lines $w + y = 0$ and $y = 0$ cut the (w, y) plane into four wedges:

$$\begin{aligned} R_1 &= \{w + y > 0, y < 0\}; \\ R_2 &= \{w + y > 0, y > 0\}; \\ R_3 &= \{w + y < 0, y < 0\}; \\ R_4 &= \{w + y < 0, y > 0\}. \end{aligned}$$

Note that

$$P(W + Y > 0, Y < 0) = P((W, Y) \in R_1). \tag{3.106}$$

Since W and Y are independent normal random variables, their joint probability density function is rotationally symmetric, and therefore

$$P((W, Y) \in R_1) = P((W, Y) \in R_4); \tag{3.107}$$
$$P((W, Y) \in R_2) = P((W, Y) \in R_3); \tag{3.108}$$
$$P((W, Y) \in R_2) = 3P((W, Y) \in R_1);$$

see Figure 3.1.

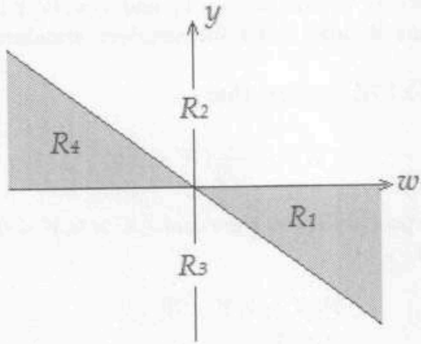

Figure 3.1: The regions R_1 to R_4 in the (w, y) plane.

Also, note that

$$\sum_{i=1}^{4} P((W,Y) \in R_i) = 1, \qquad (3.109)$$

since $P(W+Y=0 \text{ or } Y=0)=0$.

From (3.107–3.109), we find that

$$\begin{aligned} 1 &= \sum_{i=1}^{4} P((W,Y) \in R_i) \\ &= 2P((W,Y) \in R_1) + 2P((W,Y) \in R_2) \\ &= 8P((W,Y) \in R_1), \end{aligned}$$

and therefore

$$P((W,Y) \in R_1) = \frac{1}{8}.$$

Thus,

$$P(W+Y>0, Y<0) = \frac{1}{8}; \qquad (3.110)$$

see (3.106).

Then, from (3.105) and (3.110), it follows that

$$P(X > 0, Y < 0) = \frac{1}{8}. \qquad (3.111)$$

From (3.102), (3.103), and (3.111), we conclude that

$$\begin{aligned} P(X > 0 | Y < 0) &= \frac{P(X > 0, Y < 0)}{P(Y < 0)} \\ &= \frac{1}{4}. \quad \square \end{aligned}$$

Question 6. If X and Y are lognormal random variables, is their product XY lognormally distributed?

Answer: First, note that, if X and Y are independent lognormal random variables, then XY is lognormally distributed, since $\ln(XY) = \ln(X) + \ln(Y)$ is the sum of two independent normal random variables, and therefore it is normally distributed.

In a more general setting, if $\ln(X)$ and $\ln(Y)$ have joint normal distribution, then $\ln(X) + \ln(Y)$ is normally distributed and therefore XY is a lognormal random variable.

Otherwise, $\ln(X) + \ln(Y)$ may not be normally distributed even if $\ln(X)$ and $\ln(Y)$ are normally distributed, in which case XY is not lognormally distributed. $\square$

Question 7. Let X be a normal random variable with mean μ and variance σ^2, and let Φ be the cumulative distribution function of the standard normal distribution. Find the expected value of $Y = \Phi(X)$.

Answer: Let Z be a standard normal random variable independent of X. Then,

$$Y = \Phi(X) = P(Z \le X | X) = E[\mathbf{1}_{Z \le X} | X],$$

and therefore

$$E[Y] \;=\; E\left[E\left[\mathbf{1}_{Z\leq X}|\,X\right]\right]. \qquad (3.112)$$

Recall from the Tower Property for conditional expectation[10] that, for any two random variables T and W,

$$E[T] \;=\; E[E[T|W]]. \qquad (3.113)$$

Using (3.113) for $T = \mathbf{1}_{Z\leq X}$ and $W = X$, we obtain that

$$\begin{aligned} E\left[E\left[\mathbf{1}_{Z\leq X}|\,X\right]\right] &= E\left[\mathbf{1}_{Z\leq X}\right] \\ &= P(Z \leq X). \end{aligned} \qquad (3.114)$$

From (3.112) and (3.114), we obtain that

$$E[Y] \;=\; P(Z \leq X). \qquad (3.115)$$

Recall that X is a normal random variable with mean μ and variance σ^2, and that X and Z are independent. Then, $Z - X$ is a normal random variable with the following mean and variance:

$$\begin{aligned} E[Z-X] &= -\mu; \\ \mathrm{var}(Z-X) &= \mathrm{var}(Z) + \mathrm{var}(X) \;=\; 1+\sigma^2. \end{aligned}$$

and $Z - X$ is a normal random variable with mean $-\mu$ and variance $1+\sigma^2$. Thus, $\frac{Z-X+\mu}{\sqrt{1+\sigma^2}}$ is a standard normal random variable and therefore

$$\begin{aligned} P(Z \leq X) &= P(Z - X \leq 0) \\ &= P\left(\frac{Z-X+\mu}{\sqrt{1+\sigma^2}} \leq \frac{\mu}{\sqrt{1+\sigma^2}}\right) \\ &= \Phi\left(\frac{\mu}{\sqrt{1+\sigma^2}}\right). \end{aligned} \qquad (3.116)$$

[10] In other words, to calculate the expected value of T, one can first calculate the conditional expected value of T knowing the extra information from W, then average out the resulting conditional expected value over W.

From (3.115) and (3.116) we conclude that

$$E[Y] = \Phi\left(\frac{\mu}{\sqrt{1+\sigma^2}}\right). \qquad (3.117)$$

We note that, if X is the standard normal variable, then $Y = \Phi(X)$ is uniformly distributed in the interval $[0,1]$, and therefore $E[Y] = \frac{1}{2}$. Note that, for $\mu = 0$ and $\sigma = 1$ in (3.117), i.e., if X is a standard normal variable, then $E[Y] = \Phi(0) = \frac{1}{2}$, which is consistent to the comment above. □

Question 8. What is the law of large numbers?

Answer: There is a strong law of large numbers and there is a weak law of large numbers. The strong law of large numbers states that the average of a large number of independent identically distributed integrable random variables converges almost surely to their common mean; in the case of the weak law of large numbers, the convergence is only in probability.

More precisely, let X_1, X_2, ... be a sequence of independent identically distributed random variables with finite expected value $\mu = E[X_i]$, and let $S_n = X_1 + \cdots + X_n$.

The strong law of large numbers states that $\frac{S_n}{n} \rightarrow \mu$ almost surely, i.e.,

$$P\left(\lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1. \qquad (3.118)$$

The weak law of large numbers states that $\frac{S_n}{n} \rightarrow \mu$ in probability, i.e.,

$$\lim_{n\to\infty} P\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) = 0, \quad \forall\, \epsilon > 0. \qquad (3.119)$$

Note that, if a sequence of random variables convergences almost surely, than it also converges in probability, and therefore, if (3.118) holds true, then (3.119) also holds true. This is the sense in which the strong law of large numbers is "stronger" than the weak law of large numbers. □

Question 9. What is the central limit theorem?

Answer: The central limit theorem states that the limiting distribution of the centered and scaled sum of an independent identically distributed sequence of random variables is a normal distribution if the common distribution of the random variables has finite variance.

More precisely, let $X_1, X_2, \ldots$ be a sequence of independent identically distributed random variables with finite expected value $\mu = E[X_i]$ and finite variance $\sigma^2 = \mathrm{var}[X_i]$. Let $S_n = X_1 + \cdots + X_n$. Then,

$$\lim_{n\to\infty} \frac{S_n - n\mu}{\sigma\sqrt{n}} = Z,$$

where Z is the standard normal distribution, and the convergence is in distribution, i.e.,

$$\lim_{n\to\infty} P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \le t\right) = P(Z \le t).$$

Putting together the Law of Large Numbers and the Central Limit Theorem, the following approximation as n goes to infinity holds:

$$\frac{S_n}{n} \approx \mu + \frac{\sigma}{\sqrt{n}} Z. \quad \square$$

Question 10. What is a martingale? How is it related to option pricing?

Answer: Let $(\Omega, \mathcal{F}_t, P)$ be a filtered probability space, where Ω is the sample space, $\mathcal{F}_t$ is a filtration, and P is a probability measure on Ω. A stochastic process X_t

is called a *martingale* with respect to the filtration $\{\mathcal{F}_t\}$ if and only if
(i) X_t is adapted, i.e., X_t is $\mathcal{F}_t$-measurable for all t;
(ii) X_t is integrable for all t, i.e., $E[|X_t|] < \infty$ for all t;
(iii) $E[X_t|\mathcal{F}_s] = X_s$ almost surely for all $s < t$.

In other words, a martingale is a stochastic process in which, given the available information $\mathcal{F}_s$ up to the current time s, the optimal estimation (in the least square sense) of the process in the future time t, i.e., $E[X_t|\mathcal{F}_s]$, is the current value X_s almost surely.

The martingale concept is one of the cornerstones of option pricing theory. The fundamental theorem of asset pricing states that, if a market model is arbitrage–free, then there exists a risk neutral probability so that each discounted asset price process under the risk neutral probability is a martingale. Thus, a way to price a derivative security is to figure out a partial differential equation, called the pricing equation, usually deduced by applying Itô's formula, so that the discounted price process of the derivative is a martingale under risk neutral probability.

Question 11. Explain the assumption $(dW_t)^2 = dt$ used in the informal derivation of Itô's Lemma.

Answer: The notation in differential form $(dW_t)^2 = dt$ is a shorthand for the conventional integral notation used in Riemann integral

$$\int_0^T (dW_t)^2 = \int_0^T dt. \tag{3.120}$$

The intuition behind (3.120) is related to the *quadratic variation* of a Brownian path. By definition, the quadratic variation $QV_W[0,T]$ of the Brownian path W in the interval $[0,T]$ is

$$QV_W[0,T] = \lim_{n\to\infty} \sum_{i=1}^{n} \left|W_{t_i} - W_{t_{i-1}}\right|^2,$$

where $t_i = \frac{i}{n}T$, for $i = 0 : n$. Therefore, in expectation, we have that

$$\begin{aligned} & E\left[\lim_{n\to\infty}\sum_{i=1}^{n}\left|W_{t_i} - W_{t_{i-1}}\right|^2\right] \\ = & \lim_{n\to\infty}\sum_{i=1}^{n} E\left[\left|W_{t_i} - W_{t_{i-1}}\right|^2\right] \\ = & \lim_{n\to\infty}\sum_{i=1}^{n}(t_i - t_{i-1}) \\ = & \ T, \end{aligned}$$

since $W_{t_i} - W_{t_{i-1}} \sim N(0, t_i - t_{i-1})$ for $i = 1 : n$. In fact, it can be shown that the convergence

$$\sum_{i=1}^{n}\left|W_{t_i} - W_{t_{i-1}}\right|^2 \longrightarrow T \text{ as } n \to \infty$$

is in L^2 sense, i.e.,

$$\lim_{n\to\infty} E\left[\left|\sum_{i=1}^{n}\left|W_{t_i} - W_{t_{i-1}}\right|^2 - T\right|^2\right] = 0.$$

By mimicking the notations used in the conventional Riemann integral, we write that

$$\begin{aligned} \int_0^T (dW_t)^2 &= \lim_{n\to\infty}\sum_{i=1}^{n}\left|W_{t_i} - W_{t_{i-1}}\right|^2 = T \\ &= \int_0^T dt, \end{aligned}$$

whose shorthand notation in differential form is

$$(dW_t)^2 = dt. \quad \square$$

Question 12. If W_t is a Wiener process, find $E[W_t W_s]$.

Answer: Assume that $s \leq t$. Write W_t as $W_t = (W_t - W_s) + W_s$, and note that

$$W_t W_s = (W_t - W_s)W_s + W_s^2. \tag{3.121}$$

Since W_τ, $\tau \geq 0$, is a Wiener process, it follows that $W_t - W_s$ and W_s are independent normal random variables of mean 0, and therefore

$$\begin{aligned} E[(W_t - W_s)W_s] &= E[W_t - W_s]E[W_s] \\ &= 0. \end{aligned} \tag{3.122}$$

Also, W_s is normal of mean $E[W_s] = 0$ and variance $\mathrm{var}(W_s) = s$. Thus,

$$\begin{aligned} \mathrm{var}(W_s) &= E[W_s^2] - (E[W_s])^2 \\ &= E[W_s^2], \end{aligned}$$

and therefore

$$E[W_s^2] = \mathrm{var}(W_s) = s. \tag{3.123}$$

From (3.121–3.123), we obtain that

$$\begin{aligned} E[W_t W_s] &= E[(W_t - W_s)W_s] + E[W_s^2] \\ &= s. \end{aligned} \tag{3.124}$$

Since (3.124) was derived under the assumption $s \leq t$, we conclude that

$$E[W_t W_s] = \min(s, t). \quad \square \tag{3.125}$$

Question 13. If W_t is a Wiener process, what is $\mathrm{var}(W_t + W_s)$?

Answer: Assume that $s \leq t$. Write W_t as $W_t = (W_t - W_s) + W_s$, and note that $W_t + W_s = (W_t - W_s) + 2W_s$.

Then,

$$\begin{aligned}
\operatorname{var}(W_t + W_s) &= \operatorname{var}(W_t - W_s) \\
&\quad + 4\operatorname{cov}(W_t - W_s, W_s) \\
&\quad + 4\operatorname{var}(W_s). \qquad (3.126)
\end{aligned}$$

Since W_τ, $\tau \geq 0$, is a Wiener process, it follows that

$$\begin{aligned}
\operatorname{cov}(W_t - W_s, W_s) &= 0; \qquad (3.127) \\
\operatorname{var}(W_t - W_s) &= t - s; \qquad (3.128) \\
\operatorname{var}(W_s) &= s, \qquad (3.129)
\end{aligned}$$

since $W_t - W_s$ and W_s are independent normal variables of variance $t - s$ and s, respectively.

From (3.126–3.129), we obtain that

$$\operatorname{var}(W_t + W_s) = (t - s) + 4s = t + 3s. \qquad (3.130)$$

Since (3.130) was derived under the assumption $s \leq t$, we conclude that

$$\operatorname{var}(W_t + W_s) = \max(s, t) + 3\min(s, t). \quad \square$$

Question 14. Let W_t be a Wiener process. Find Find

$$\int_0^t W_s \, dW_s \ \text{ and } \ E\left[\int_0^t W_s \, dW_s\right].$$

Answer: Since $\int x \, dx = \frac{x^2}{2} + C$, we begin by computing $d\left(\frac{W_t^2}{2}\right)$. Recall from Itô's lemma that, if $f(x,t)$ is a continuously differentiable function, then

$$df = \frac{\partial f}{\partial t} \, dt + \frac{\partial f}{\partial x} \, dW_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2} \, dt.$$

For $f(x,t) = \frac{x^2}{2}$, we find that

$$d\left(\frac{W_t^2}{2}\right) = W_t dW_t + \frac{1}{2} dt,$$

and therefore

$$d\left(\frac{W_t^2 - t}{2}\right) = W_t dW_t. \qquad (3.131)$$

By integrating (3.131) between 0 and t, and since $W_0 = 0$, we obtain that

$$\int_0^t W_s\, dW_s = \frac{W_t^2 - t}{2}, \ \forall\, t \geq 0. \qquad (3.132)$$

From (3.132), it follows that

$$E\left[\int_0^t W_s\, dW_s\right] = \frac{E[W_t^2] - t}{2} = 0,$$

since W_t is normally distributed with mean 0 and variance t and therefore

$$\begin{aligned} E[W_t^2] &= \operatorname{var}(W_t) + (E[W_t])^2 \\ &= t. \quad \square \end{aligned}$$

Question 15. Find the distribution of the random variable

$$X = \int_0^1 W_t dW_t.$$

Answer: Recall from Itô's formula that, if W_t is a Wiener process and $f(x)$ is a function with continuous second order derivative, then

$$df(W_t) = f'(W_t)dW_t + \frac{1}{2}f''(W_t)dt. \qquad (3.133)$$

For $f(x) = x^2$, we obtain from (3.133) that

$$dW_t^2 = 2W_t dW_t + dt. \qquad (3.134)$$

By integrating (3.134) from 0 to 1, we obtain that

$$W_1^2 \;=\; 2\int_0^1 W_t dW_t + 1 \;=\; 2X + 1. \qquad (3.135)$$

Note that W_1 is a standard normal random variable, and let $W_1 = Z$. By solving (3.135) for X, we find that

$$X \;=\; \frac{W_1^2 - 1}{2} \;=\; \frac{Z^2 - 1}{2}.$$

Let $f_X(x)$ and $F_X(x)$ be the probability density function of X and the cumulative distribution function of X, respectively.

Note that

$$P\left(X \leq -\frac{1}{2}\right) \;=\; P(Z^2 \leq 0) \;=\; 0.$$

Thus, if $x < -\frac{1}{2}$, then $F_X(x) = P(X \leq x) = 0$ and therefore

$$f_X(x) \;=\; 0,$$

since $f_X(x) = F_X'(x)$.

If $x \geq -\frac{1}{2}$, then

$$\begin{aligned}
F_X(x) &= P(X \leq x) = P\left(\frac{Z^2 - 1}{2} \leq x\right) \\
&= P\left(Z^2 \leq 2x + 1\right) \\
&= P\left(-\sqrt{2x+1} \leq Z \leq \sqrt{2x+1}\right) \\
&= 2\,P\left(0 \leq Z \leq \sqrt{2x+1}\right) \\
&= \frac{2}{\sqrt{2\pi}} \int_0^{\sqrt{2x+1}} e^{-\frac{y^2}{2}}\, dy.
\end{aligned}$$

By differentiating $F_X(x)$, it follows that, for $x < -\frac{1}{2}$,

$$\begin{aligned} f_X(x) &= F_X'(x) \\ &= \sqrt{\frac{2}{\pi}} \frac{d}{dx} \left(\int_0^{\sqrt{2x+1}} e^{-\frac{y^2}{2}} dy \right) \\ &= \sqrt{\frac{2}{\pi}} e^{-\frac{(\sqrt{2x+1})^2}{2}} \left(\sqrt{2x+1}\right)' \\ &= \sqrt{\frac{2}{\pi}} \frac{e^{-\frac{2x+1}{2}}}{\sqrt{2x+1}}. \end{aligned}$$

We conclude that

$$f_X(x) = \begin{cases} \sqrt{\frac{2}{\pi}} \frac{e^{-\frac{2x+1}{2}}}{\sqrt{2x+1}}, & \text{if } x > -\frac{1}{2}; \\ 0 & \text{if } x \le -\frac{1}{2}. \end{cases} \qquad \square$$

Question 16. Let W_t be a Wiener process. Find the mean and the variance of

$$\int_0^t W_s^2 dW_s.$$

Answer: Let $\mathcal{B}_t$ be the Borel σ-algebra over the time interval $[0, t]$ and let $\mathcal{F}_t$ be the filtration for the probability space in which the Wiener process W_t resides. We will use the following results:[11]

Theorem 3.1. *(Martingality)*
Let f_s be progressively measurable and square integrable in $[0, T]$, i.e., f_t is $\mathcal{B}_t \otimes \mathcal{F}_t$-measurable for every $t \in [0, T]$ and $E\left[\int_0^T |f_t|^2 dt\right] < \infty$. Then, the stochastic integral

[11] For the proofs of Theorem 3.1 and Theorem 3.2, see Theorem 2.8 on p.65 and Theorem 3.1 on p.67, respectively, from Friedman [1].

$\int_0^t f_s dW_s$ defines a zero mean, square integrable martingale for $t \in [0,T]$. In particular,

$$E\left[\int_0^t f_s dW_s\right] = 0, \quad \forall\, t \in [0,T]. \tag{3.136}$$

Theorem 3.2. *(Itô's isometry)*
Let f_t and g_t be progressively measurable and square integrable processes. Then,

$$E\left[\int_0^t f_s dW_s \int_0^t g_s dW_s\right] = \int_0^t E[f_s\, g_s] ds.$$

In particular, if $f_s = g_s$, it follows that

$$E\left[\left(\int_0^t f_s dW_s\right)^2\right] = \int_0^t E[f_s^2] ds. \tag{3.137}$$

For our problem, we need to compute $E[X]$ and $\mathrm{var}(X)$, where

$$X \;=\; \int_0^t W_s^2 dW_s.$$

We first check that the integrand W_s^2 is progressively measurable and square integrable in $[0,t]$.

Note that W_s^2 is progressively measurable because it is adapted and continuous.

Furthermore, since $W_s \sim N(0,s)$, it follows that $W_s = \sqrt{s}Z$, where Z is a standard normal random variable, and therefore

$$E[W_s^4] = E\left[\left(\sqrt{s}Z\right)^4\right] = s^2\, E[Z^4] = 3s^2, \tag{3.138}$$

where we used the fact that the fourth moment of the standard normal distribution is 3, i.e., $E[Z^4] = 3$. From (3.138), it follows that

$$\int_0^t E[W_s^4] ds = \int_0^t 3s^2 ds = t^3 < \infty. \tag{3.139}$$

In other words, W_s^2 is square integrable in $[0, t]$.

We can therefore apply both Theorems 3.1 and 3.2 with $f_s = W_s^2$ and $g_s = W_s^2$.

From (3.136) for $f_s = W_s^2$, we find that

$$E[X] = E\left[\int_0^t W_s^2 dW_s\right] = 0. \qquad (3.140)$$

From (3.137) for $f_s = g_s = W_s^2$, and using (3.139), it follows that

$$\begin{aligned} E[X^2] &= E\left[\left(\int_0^t W_s^2 dW_s\right)^2\right] = \int_0^t E[W_s^4] ds \\ &= t^3. \end{aligned}$$

Since $E[X] = 0$, see (3.140), we find that

$$\operatorname{var}(X) = E[X^2] - (E[X])^2 = t^3. \quad \square$$

Question 17. If W_t is a Wiener process, find the variance of

$$X = \int_0^1 \sqrt{t} e^{\frac{W_t^2}{8}} dW_t.$$

Answer: We will use Theorems 3.1 and 3.2 to solve this problem. To do so, we first check that the integrand $\sqrt{t} e^{\frac{W_t^2}{8}}$ is progressively measurable and square integrable in $[0, 1]$.

The process $\sqrt{t} e^{\frac{W_t^2}{8}}$ is progressively measurable because it is adapted and continuous.

Furthermore, since $W_t \sim N(0,t)$, the probability density function of W_t is $\frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$, and therefore

$$\begin{aligned}
E\left[e^{\frac{W_t^2}{4}}\right] &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} e^{\frac{x^2}{4}} e^{-\frac{x^2}{2t}}\, dx \\
&= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} e^{\left(\frac{1}{4}-\frac{1}{2t}\right)x^2}\, dx \\
&= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\cdot\frac{2-t}{2t}x^2}\, dx \\
&= \frac{1}{\sqrt{2\pi t}} \sqrt{2\pi \cdot \frac{2t}{2-t}} \qquad (3.141) \\
&= \sqrt{\frac{2}{2-t}}, \qquad (3.142)
\end{aligned}$$

where, for (3.141), we used the identity

$$\int_{-\infty}^{\infty} e^{-\frac{Ax^2}{2}}\, dx = \sqrt{\frac{2\pi}{A}},$$

for any positive constant $A > 0$. From (3.142), it follows that

$$\begin{aligned}
\int_0^1 E\left[te^{\frac{W_t^2}{4}}\right] dt &= \int_0^1 t\, E\left[e^{\frac{W_t^2}{4}}\right] dt \\
&= \int_0^1 t \cdot \sqrt{\frac{2}{2-t}} dt \\
&= -\frac{2\sqrt{2}}{3} \left((t+4)\sqrt{2-t}\right)\Big|_0^1 \\
&= \frac{2}{3}\left(8 - 5\sqrt{2}\right). \qquad (3.143)
\end{aligned}$$

Thus, $\int_0^1 E\left[te^{\frac{W_t^2}{4}}\right] dt < \infty$, and we conclude that both Theorems 3.1 and 3.2 can be applied here.

From (3.136), it follows that

$$E[X] = E\left[\int_0^1 \sqrt{t} e^{\frac{W_t^2}{8}}\, dW_t\right] = 0. \qquad (3.144)$$

From (3.137), and using (3.143), we obtain that

$$\begin{aligned} E[X^2] &= E\left[\left(\int_0^1 \sqrt{t}e^{\frac{W_t^2}{8}}\,dW_t\right)^2\right] \\ &= \int_0^1 E\left[\left(\sqrt{t}e^{\frac{W_t^2}{8}}\right)^2\right] dt \\ &= \int_0^1 E\left[te^{\frac{W_t^2}{4}}\right] dt \\ &= \frac{2}{3}(8-5\sqrt{2}). \end{aligned} \tag{3.145}$$

From (3.144) and (3.145), we conclude that

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = \frac{2}{3}\left(8-5\sqrt{2}\right). \quad \square$$

Question 18. If W_t is a Wiener process, what is $E\left[e^{W_t}\right]$?

Answer:

Solution 1: If W_t is a Wiener process and $Y = e^{W_t}$, then

$$\ln(Y) = W_t \cong \sqrt{t}Z, \tag{3.146}$$

where Z is the standard normal random variable, and therefore Y is a lognormal random variable. Recall that the expected value of a lognormal random variable given by $\ln(V) = \mu + \sigma Z$ is

$$E[V] = e^{\mu + \frac{\sigma^2}{2}}. \tag{3.147}$$

From (3.146), and using (3.147) with $\mu = 0$ and $\sigma = \sqrt{t}$, we conclude that

$$E\left[e^{W_t}\right] = E[Y] = e^{\frac{t}{2}}.$$

Solution 2: Since $W_t \cong \sqrt{t}Z$, it follows that

$$\begin{aligned} E\left[e^{W_t}\right] &= E\left[e^{\sqrt{t}Z}\right] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{\sqrt{t}x}e^{-\frac{x^2}{2}}\,dx \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{\sqrt{t}x-\frac{x^2}{2}}\,dx. \qquad (3.148) \end{aligned}$$

By completing the square, we find that

$$\sqrt{t}x - \frac{x^2}{2} = -\frac{1}{2}\left(x-\sqrt{t}\right)^2 + \frac{t}{2},$$

and therefore

$$e^{\sqrt{t}x-\frac{x^2}{2}} = e^{\frac{t}{2}}e^{-\frac{1}{2}\left(x-\sqrt{t}\right)^2}. \qquad (3.149)$$

From (3.148) and (3.149), it follows that

$$\begin{aligned} E\left[e^{W_t}\right] &= e^{\frac{t}{2}}\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(x-\sqrt{t}\right)^2}\,dx \\ &= e^{\frac{t}{2}}\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\,dy \qquad (3.150) \\ &= e^{\frac{t}{2}}, \qquad (3.151) \end{aligned}$$

where we used the substitution $y = x - \sqrt{t}$ for (3.150), and, for (3.151), we used the fact that

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\,dy = 1,$$

since $\frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}}$ is the probability density function of the standard normal distribution. □

Question 19. If W_t is a Wiener process, find the variance of

$$\int_0^t s\,dW_s.$$

Answer: Recall that, for any deterministic square integrable function $f : [a,b] \to \mathbb{R}$, the stochastic integral $\int_a^b f(s)dW_s$ is normally distributed with mean 0 and variance equal to the square of the L^2 norm of f, i.e.,

$$\int_a^b f(s)dW_s \sim N\left(0, \int_a^b f^2(s)ds\right).$$

Then,

$$\int_0^t s\,dW_s \;\sim\; N\left(0, \int_0^t s^2 ds\right) \;=\; N\left(0, \frac{t^3}{3}\right),$$

and we conclude that

$$\mathrm{var}\left(\int_0^t s dW_s\right) = \frac{t^3}{3}. \quad \square$$

Question 20. Let W_t be a Wiener process, and let

$$X_t \;=\; \int_0^t W_\tau d\tau. \tag{3.152}$$

What is the distribution of X_t? Is X_t a martingale?

Answer: A solution to this question was given in Chapter 1 using integration by parts; we include a different solution here.

Note that X_t is not a martingale because, if we rewrite (3.152) in differential form as

$$dX_t \;=\; W_t dt \;=\; W_t dt + 0\,dW_t,$$

we can think of X_t as being a diffusion process with only the drift part W_t.

Recall that the integral of a one-parameter family of Gaussian random variables remains Gaussian. Since W_t is a Gaussian family, X_t is normally distributed. Furthermore,

$$E[X_t] = \int_0^t E[W_\tau]d\tau = 0.$$

Therefore,

$$\mathrm{var}(X_t) \;=\; E[X_t^2] - (E[X_t])^2 \;=\; E[X_t^2]. \qquad (3.153)$$

Note that

$$X_t^2 \;=\; \left(\int_0^t W_s ds \right)^2 = \int_0^t \int_0^t W_s W_u ds du, \qquad (3.154)$$

and recall that

$$E[W_s W_u] \;=\; \min\{s,u\}, \;\; \forall\, s,u > 0; \qquad (3.155)$$

see (3.125).

From (3.153–3.155), we obtain that

$$\begin{aligned}
\mathrm{var}(X_t) \;&=\; E[X_t^2] \;=\; \int_0^t \int_0^t E[W_s W_u] ds du \\
&=\; \int_0^t \int_0^t \min\{s,u\} ds du \\
&=\; \int_0^t \left(\int_0^u s ds + \int_u^t u ds \right) du \\
&=\; \int_0^t \left(\frac{u^2}{2} + u(t-u) \right) du \\
&=\; \int_0^t \left(ut - \frac{u^2}{2} \right) du \\
&=\; \frac{t^3}{2} - \frac{t^3}{6} \\
&=\; \frac{t^3}{3}.
\end{aligned}$$

We conclude that X_t is normally distributed with mean 0 and variance $\frac{t^3}{3}$, i.e.,

$$X_t \;\sim\; N\left(0, \frac{t^3}{3}\right). \quad \square$$

Question 21. What is an Itô process?

Answer: An Itô process is a generic term referring to a stochastic process X_t determined by the solution of a stochastic differential equation (SDE) of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \qquad (3.156)$$

where W_t is a Wiener process. The coefficient $a(x,t)$ of the dt term is the drift of X_t; the coefficient $b(x,t)$ of the dW_t term is the diffusion of X_t. The SDE (3.156) is, by definition, the shorthand notation for the stochastic integral equation

$$X_t = X_0 + \int_0^t a(X_s, s)\, ds + \int_0^t b(X_s, s)\, dW_s.$$

We note that a sufficient condition for the existence and uniqueness of the (strong) solution to an SDE is for the drift and the diffusion coefficients $a(x,t)$ and $b(x,t)$ to be locally Lipschitz functions of at most linear growth in x. □

Question 22. What is Itô's lemma?

Answer: Itô's lemma, also known as Itô's formula, states that, if X_t is an Itô process satisfying the SDE

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t,$$

then for any function $f(x,t)$ with continuous second order partial derivative in x and continuous first order partial derivative in t, the process $f(X_t, t)$ is also an Itô process, driven by the same Wiener process W_t, and the drift and diffusion parts of $f(X_t, t)$ are determined according to the Taylor expansion of $f(x,t)$ to first order for the t part and up to second order for the x part:

$$\begin{aligned}
df(X_t,t) &= \frac{\partial f}{\partial t}\,dt + \frac{\partial f}{\partial x}\,dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\,(dX_t)^2 \\
&= \frac{\partial f}{\partial t}\,dt + \frac{\partial f}{\partial x}\,[a(X_t,t)dt + b(X_t,t)dW_t] \\
&\quad + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\,[a(X_t,t)dt + b(X_t,t)dW_t]^2 \\
&= \left[\frac{\partial f}{\partial t} + a(X_t,t)\frac{\partial f}{\partial x} + \frac{b^2(X_t,t)}{2}\frac{\partial^2 f}{\partial x^2}\right]\,dt \\
&\quad + b(X_t,t)\frac{\partial f}{\partial x}\,dW_t. \qquad (3.157)
\end{aligned}$$

Note that for (3.157) we used the fact that

$$\begin{aligned}
(dX_t)^2 &= [a(X_t,t)dt + b(X_t,t)dW_t]^2 \\
&= b^2(X_t,t)dt
\end{aligned}$$

since $(dW_t)^2 = dt$ (see Problem 3.6 in this section), $(dt)^2 = 0$, and $dW_t\,dt = 0$. $\square$

Question 23. If W_t is a Wiener process, is the process $X_t = W_t^2$ a martingale?

Answer: Recall that a stochastic process M_t defined in a filtered probability space $(\Omega, \mathcal{F}_t, P)$ is a martingale if and only if
(i) M_t is adapted, i.e., M_t is $\mathcal{F}_t$-measurable for all t;
(ii) M_t is integrable for all t, i.e., $E[M_t] < \infty$ for all t;
(iii) $E[|M_t||\mathcal{F}_s] = M_s$ almost surely for all $s < t$.

We check whether the process X_t satisfies the conditions (i), (ii), and (iii) above.

Since any continuous function of the Wiener process W_t is adapted, the process X_t is adapted, and therefore condition (i) is satisfied.

Also, $E[X_t] = E[W_t^2] = t < \infty$, since W_t is a normal random variable of mean 0 and variance t, and therefore

$E[W_t^2] = \mathrm{var}[W_t] = t$. Thus, the random variable X_t is integrable for every t, and X_t satisfies condition (ii).

To check whether X_t satisfies condition (iii), note that, for every $s < t$,

$$\begin{aligned} E\left[X_t|\mathcal{F}_s\right] &= E\left[W_t^2|\mathcal{F}_s\right] \\ &= E\left[(W_t - W_s + W_s)^2|\mathcal{F}_s\right] \\ &= E\left[(W_t - W_s)^2|\mathcal{F}_s\right] + 2E\left[(W_t - W_s)W_s|\mathcal{F}_s\right] \\ &\quad + E\left[W_s^2|\mathcal{F}_s\right]. \end{aligned} \tag{3.158}$$

Since the Wiener process W_t has independent increments, i.e, $W_t - W_s$ is independent of $\mathcal{F}_s$, and stationary increment, i.e., $W_t - W_s \sim W_{t-s}$. Moreover, $W_t - W_s$ is a normal random variable of mean 0 and variance $t-s$, i.e., $E[W_{t-s}] = 0$ and $\mathrm{var}(W_{t-s}) = E[W_{t-s}^2] = t - s$. Then,

$$E\left[(W_t - W_s)^2|\mathcal{F}_s\right] = E[W_{t-s}^2] = t - s, \tag{3.159}$$

and

$$\begin{aligned} E\left[(W_t - W_s)W_s|\mathcal{F}_s\right] &= W_s E\left[W_t - W_s|\mathcal{F}_s\right] \\ &= W_s E[W_{t-s}] \\ &= 0. \end{aligned} \tag{3.160}$$

Since W_s is $\mathcal{F}_s$-measurable, it follows that

$$E[W_s^2|\mathcal{F}_s] = W_s^2. \tag{3.161}$$

From (3.158–3.161), we find that

$$\begin{aligned} & E\left[X_t|\mathcal{F}_s\right] \\ = \;& E\left[(W_t - W_s)^2|\mathcal{F}_s\right] + 2E\left[(W_t - W_s)W_s|\mathcal{F}_s\right] \\ & + E\left[W_s^2|\mathcal{F}_s\right] \\ = \;& t - s + W_s^2. \end{aligned}$$

Thus,

$$E\left[X_t|\mathcal{F}_s\right] = t - s + W_s^2 \neq W_s^2 = X_s,$$

and therefore the process X_t does not satisfy condition (iii).

We conclude that X_t is not a martingale. $\Box$

Question 24. If W_t is a Wiener process, is the process

$$N_t = W_t^3 - 3tW_t$$

a martingale?

Answer:

Solution 1: A stochastic process M_t defined in a filtered probability space $(\Omega, \mathcal{F}_t, P)$ is called a martingale if and only if

(i) M_t is adapted, i.e., M_t is $\mathcal{F}_t$-measurable for all t;
(ii) M_t is integrable for all t, i.e., $E[|M_t|] < \infty$ for all t;
(iii) $E[M_t|\mathcal{F}_s] = M_s$ almost surely for all $s < t$.

We check whether the process N_t satisfies the conditions (i), (ii), and (iii) above.

Since any continuous function of the Wiener process W_t is adapted, the process N_t is adapted, and therefore condition (i) is satisfied.

Also,

$$E[N_t] = E[W_t^3] - 3tE[W_t] = 0 < \infty,$$

since $W_t \sim N(0,t)$, therefore $E[W_t] = E[W_t^3] = 0$. Thus, the random variable N_t is integrable for every t, and N_t satisfies condition (ii).

To check whether N_t satisfies condition (iii), note that, for every $s < t$,

$$\begin{aligned}
& E\left[N_t|\mathcal{F}_s\right] \\
= \; & E\left[W_t^3 - 3tW_t|\mathcal{F}_s\right] \\
= \; & E\left[(W_t - W_s + W_s)^3|\mathcal{F}_s\right] - 3tE\left[W_t|\mathcal{F}_s\right] \\
= \; & E\left[(W_t - W_s)^3|\mathcal{F}_s\right] + 3E\left[(W_t - W_s)^2 W_s|\mathcal{F}_s\right] \\
& + \; 3E\left[(W_t - W_s)W_s^2|\mathcal{F}_s\right] \; + \; E\left[W_s^3|\mathcal{F}_s\right] \\
& - \; 3tE\left[W_t|\mathcal{F}_s\right].
\end{aligned}$$

Recall that the Wiener process W_t has independent increments, i.e, $W_t - W_s$ is independent of $\mathcal{F}_s$, and stationary increment, i.e., $W_t - W_s \sim W_{t-s}$, and the fact that W_s is $\mathcal{F}_s$-measurable. Then,

$$\begin{aligned}
E\left[(W_t - W_s)^3|\mathcal{F}_s\right] &= E[W_{t-s}^3] \\
&= 0; \qquad (3.162) \\
E\left[(W_t - W_s)^2 W_s|\mathcal{F}_s\right] &= W_s\ E[W_{t-s}^2] \\
&= (t-s)W_s; \qquad (3.163) \\
E\left[(W_t - W_s)W_s^2|\mathcal{F}_s\right] &= W_s^2\ E[W_{t-s}] \\
&= 0; \qquad (3.164) \\
E\left[W_s^3|\mathcal{F}_s\right] &= W_s^3, \qquad (3.165)
\end{aligned}$$

since $W_{t-s} \sim N(0, t-s)$, and therefore

$$E[W_{t-s}] = E[W_{t-s}^3] = 0;$$

$$E[W_{t-s}^2] = \mathrm{var}[W_{t-s}] = t - s.$$

Moreover, since W_t is a martingale, it follows that

$$E[W_t|\mathcal{F}_s] = W_s. \qquad (3.166)$$

From (3.162–3.166), we find that

$$\begin{aligned}
& E\left[N_t|\mathcal{F}_s\right] \\
= \ & E\left[(W_t - W_s)^3|\mathcal{F}_s\right] + 3E\left[(W_t - W_s)^2 W_s|\mathcal{F}_s\right] \\
& + \ 3E\left[(W_t - W_s)W_s^2|\mathcal{F}_s\right] \ + \ E\left[W_s^3|\mathcal{F}_s\right] \\
& - \ 3tE\left[W_t|\mathcal{F}_s\right] \\
= \ & 0 + 3(t-s)W_s + 0 + W_s^3 - 3tW_s \\
= \ & W_s^3 - 3sW_s \\
= \ & N_s.
\end{aligned}$$

Thus, the process N_t satisfies condition (iii), and we conclude that N_t is a martingale.

Solution 2: By applying Itô's formula to N_t, we obtain that

$$\begin{aligned} dN_t &= d(W_t^3 - 3tW_t) \\ &= 3W_t^2 dW_t + \frac{1}{2} \cdot 6W_t dt \; - \; 3(W_t dt + t dW_t) \\ &= 3(W_t^2 - t) dW_t. \end{aligned}$$

The process N_t has zero drift and can be written as a stochastic integral as

$$N_t = \int_0^t 3(W_s^2 - s) dW_s.$$

Moreover,

$$\begin{aligned} & \int_0^t E\left[\left(3(W_s^2 - s)\right)^2\right] ds \\ = & \; 9\left[\int_0^t \left(E[W_s^4] - 2sE[W_s^2] + s^2\right) ds\right] \\ = & \; 9\left[\int_0^t \left(3s^2 - 2s^2 + s^2\right) ds\right] \qquad (3.167) \\ = & \; 6t^3 < \infty; \end{aligned}$$

for (3.167), we used the fact that $W_s = \sqrt{s}Z$, where Z is a standard normal random variable, and therefore

$$\begin{aligned} E[W_s^2] &= E\left[\left(\sqrt{s}Z\right)^2\right] \; = \; sE[Z^2] \; = \; s; \\ E[W_s^4] &= E\left[\left(\sqrt{s}Z\right)^4\right] \; = \; s^2 E[Z^4] \; = \; 3s^2. \end{aligned}$$

Therefore, the process $3(W_s^2 - s)$ is square integrable for $s \in [0, t]$, and, from Theorem 3.1, we conclude that N_t is a martingale.

Question 25. What is Girsanov's theorem?